Batteries

Though battery packs have become smaller, they are still one of the largest components of a mobile device. So far, this has been as unavoidable a situation as the display problems. All devices need power for at least eight hours. In fact, the norm today is several days of standby time and almost half a day of continuous usage capacity. In order to increase battery life, new technologies are being looked into to replace standard Lithium-ion, Nickel-Cadmium and Nickel metal hydride batteries.

The most notable of these technological advances is perhaps in the area of fuel cells.

Fuel Cells

Fuel cells use chemical reactions to "burn" a fuel, oxidise it and create electricity. Normally, fuel cells have cartridges that contain the fuel, and running out of power means exchanging the spent cartridge with a new one. For mobile devices, fuel cells are a great alternative, or even a great backup system. Imagine you're on the road, travelling, and your mobile battery dies. There's not a power outlet for miles, and you are expecting urgent calls. With a standard mobile battery, you're pretty much out of luck—even with the innovative "human-powered" chargers that are available, you don't really want to be huffing and puffing while on an important call. If your mobile was fuel cell-powered, all you would need to do would be to pop in a tiny fuel cartridges.

Fuel cells have reached a stage where they can power a standard mobile device for up to 10 times longer. So if your normal battery runs out in say 3 days, you can go a month with a fuel cell cartridge!

Recently, Motorola made an investment in Tekion Incorporated (www.tekion.com) that's being viewed as a strategic investment on the former's part. Tekion has developed micro fuel cells which they call Formira cells, since they use a purified formic acid as their fuel. These cells are hybrids that consist of advanced mobile cells (the